Architectural parameters of Embedded Systems (With Special Focus on NVIDIA Jetson)

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Acronyms

- SoC = System-on-chip
- HPC = High Performance Computing
- DSP = Digital Signal Processor
- BW = Bandwidth
- LPDDR4 = Low Power DDR4
- CUDA = Compute Unified Device Architecture
- FPGA = Field Programmable Gate Array
- cuDNN = CUDA Deep Neural Network
- IoT = Internet of Things
- TPU = Tensor Processing Unit
- RISC = Reduced Instruction Set Computer

Architectural Parameters

	Jetson Nano	TX2 (8GB)	Xavier NX (16GB)	AGX Xavier (64GB)	Raspberry Pi 3(B+)
Peak perf.	472 GFLOP	1.33 TFLOP	21 TOPS	32 TOPS	6 GFLOP (FP64)
GPU	128-core Maxwell GPU	256-core Pascal GPU	384-core Volta GPU+48 Tensor Cores	512-core Volta GPU+64 Tensor cores	VideoCore IV
CPU	ARM CPU	ARM CPU	ARM CPU	ARM CPU	ARM CPU
Memory	4 GB 64-bit LPDDR4, 25.6 GB/s	8 GB 128-bit LPDDR4, 59.7GB/s	16 GB 128-bit LPDDR4x, 59.7GB/s	64 GB 256-bit LPDDR4x, 136.5GB/s	1GB LPDDR2, 8.5GB/s
Storage (eMMC)	16GB	32GB	16GB	32GB	MicroSDHC slot
Power	5-10W	7.5-15W	10-20W	10-30W	1.5W to 6.7W
Weight (gram)	~100	~100	?	?	50

TABLE I

HARDWARE SPECIFICATION COMPARISONS ON PEAK PERFORMANCE FOR DATA TYPES FP32 AND INT8, ON-CHIP MEMORY SIZE, OFF-CHIP BANDWIDTH (BW), TDP AMONG AMD FPGA U250, NVIDIA GPU A10G, NVIDIA GPU JETSON AGX ORIN, AND AMD VERSAL ACAP VCK190

Hardware Spec.	Tech. Node	FP32	INT8	Off-chip BW	On-chip Mem.	Off-chip Mem.	TDP
AMD FPGA U250	16nm	1.2T	6.95T	77GB/s	53MB	16GB	225W
Nvidia GPU A10G	8nm	35T	140T	600GB/s	14MB	24GB	300W
Nvidia GPU Jetson Orin	8nm	5.3T	85T	204GB/s	6MB	64GB	60W
AMD ACAP VCK190	7nm	6.4T	102T	25GB/s	33MB	8GB	<180W

Unique features of Jetson

- Has both CPU and GPU: self-boot and high-performance.
- DNNs trained on high-end GPUs are coded using CUDA, they can be directly ported to Jetson for inference.
- TX1 and TX2 support cuDNN library which provides optimized implementation of important kernels.
- For other devices such as Raspberry Pi, DL kernels need to developed using OpenCL, since a cuDNN-like library is not available for them.

Architectural Parameters of Embedded Systems

	GTX TITAN X	Jetson TX2	Movidius
Processing Unit	3072 CUDA cores	256 CUDA cores	Myriad 2 VPU
FLOPS	6.7T FP32	1.33T FP32	2T FP16
Memory	12GByte GDDR5	8GByte LPDDR4	4GBit LPDDR3
Mem. Bandwidth	336.6 GBytes/s	59.7 GBytes/s	4 GBits/s
Power	250 W	15 W	1 W

Architectural Parameters of Embedded Systems

	GPU	DSP	RISC core (CPU)	FPGA
Platform	**Jetson TX1**	**66AK2H12**	**Parallella**	**ZC706**
Vendor/SoC	NVIDIA Tegra X1	TI Keystone II	Adapteva Epiphany-III	Xilinx Zynq Z7045
Technology	20nm	28nm	65nm	28nm
Processor	ARMv8+NEON	ARMv7+NEON	ARMv7 (Zynq)	ARMv7+NEON
Accelerator	Maxwell GPU 256-core	C66 DSP 8-core	Epiphany-III 16-core	Kintex FPGA 32 lanes
Host Clock	1.9GHz	1.4GHz	667MHz	667MHz CPU
Accelerator Clock	1GHz	1.2–1.4GHz	667MHz	180MHz FPGA[1]
Onchip Host Mem	64KB L1 + 2,048KB L2	32KB L1 + 1MB L2	32KB L1 + 512KB L2	32KB L1 + 512KB L2
Onchip Accel. Mem	64KB[3]	6MB MSMC	32KB/core	64KB SRAM[2]
Off-chip Memory	4GB 32b LPDDR4-800	2GB 72b DDR3-1600 2×	1GB 32b DDR3-1066	2GB 32b DDR3-1066
System Power	6W (Idle), 10W	11W (Idle), 14W	3W (Idle), 4W	17W (Idle), 19W
Host OS	Ubuntu 14.04	Bare-Metal (no OS)	Ubuntu	Bare-Metal (no OS)
Compiler +	gcc-4.8.4, nvcc 7.0	TI CCSv6, DSPLibv3.1.0.0	e-gcc	gcc-4.6.3
Library	cuDNN v4	IMGLib v3.1.1.0		Vivado 2014.2
Cost[4] (USD)	$599	$997 ($667/chip)	$126	$2,275 ($1,596/chip)

Embedded vs HPC Systems

Category	IoT/Edge Devices	GPU-Based Edge Devices		Custom-ASIC Edge Accelerators		FPGA Based	HPC Platforms CPU	HPC Platforms GPU		
Platform	Raspberry Pi 3B [34]⋆	Jetson TX2 [69]	Jetson Nano [36]	EdgeTPU [35]	Movidius NCS [37]♦	PYNQ-Z1 [64]	Xeon	RTX 2080	GTX Titan X	Titan Xp
CPU	4-core Ctx.A53 @1.2 GHz*	4-core Ctx.A57 2-core Denver2 @2 GHz	4-core Ctx.A57 @1.43 GHz	4-core Ctx.A53 & Ctx.-M4 @1.5 GHz	N/Ap	4-core Ctx.A9 @650 MHz	2x 22-core E5-2696 v4 @2.20GHz	N/Ap*	N/Ap	N/Ap
GPU	No GPGPU	256-core Pascal μA	128-core Maxwell μA	N/Ap	N/Ap	N/Ap	N/Ap	2944-core Turing μA	3072-core Maxwell μA	3840-core Pascal μA
Accelerator	N/Ap	N/Ap	N/Ap	EdgeTPU	Myriad 2 VPU	ZYNQ XC7Z020	N/Ap	N/Ap	N/Ap	N/Ap
Memory†	1 GB LPDDR2	8 GB LPDDR4	4 GB LPDDR4	N/Av*	N/Av	630 KB BRAM 512 MB DDR3	264 GB DDR4	8 GB GDDR6	12 GB GDDR5	12 GB GDDR5X
Idle Power‡	1.33	1.90	1.25	3.24	0.36	2.65	≈70	≈39	≈15	≈55
Average Power‡	2.73	9.65	4.58	4.14	1.52	5.24	300 TDP	≈	≈100	≈
Platform	All	All	All	TFLite	NCSDK	TVM/FINN	All	All	All	All

† Effective memory size used for acceleration/execution of DNNs, e.g., GPU/CPU/Accelerator memory size. * Ctx.: Arm Cortex. N/Ap: Not applicable. N/Av: Not available.
‡ : Measured idle and average power while executing DNNs, in Watts. ⋆ : Raspberry Pi 4B [70], with 4-core Ctx.A72 and maximum of 4 GB LPDDR4, was released after this paper acceptance. With better memory technology and out-of-order execution, Raspberry Pi 4B is expected to perform better. ♦ Intel Neural Compute Stick 2 [61] with a new VPU chip and support for several frameworks was announced during paper submission, but the product was not released.

Theoretical peak single-precision performance and memory bandwidth of selected mobile chips

SOC Name	Year	BW (GB/s)	Performance in GFlops		
			CPU	*GPU*	Overall
Apple A8	2014	12.8	44.8	115.2	160
Apple A9	2015	25.6	88.8	172.8	261.6
Apple A10	2016	25.6	112.3	384.6	495
Snapdragon 410	2014	8.5	19.2	10.8	30
Snapdragon 616	2015	12.8	35.2	29.7	129.8
Snapdragon 821	2016	29.9	37.4	260.0	297.4
Snapdragon 835	2017	29.9	78.4	576.0	654.4

Lan et al., FeatherCNN: Fast Inference Computation with TensorGEMM on ARM Architectures, TPDS 2019

References

https://developer.nvidia.com/embedded/jetson-modules